

THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DARESSALAAM REGION
FORM FOUR - MOCK EXAMINATION - 2026

CHEMISTRY 2A
(ACTUAL PRACTICAL)

CODE:032/2A

TIME:2:30Hours

Thursdat 21st May, 2026 A.M

INSTRUCTIONS

- i. This paper consist of two questions. Answer **ALL** questions.
- ii. Each question carries twenty five (25) marks making a total of 50 marks.
- iii. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
- iv. Write your **Examination Number** on every page of your answer booklet.
- v. You may use the following constants:
Atomic masses C = 12, O = 16, H = 1, K = 39, Na = 23, N = 14
1 dm³ = 1L = 1000cm³

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This paper consists of 2 printed pages

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1. You are provided with
 - 1.1 Solution O made by diluting 200cm^3 of 2M acetic acid (CH_3COOH) to make 2 litres of solution.
 - 1.2 Solution R made by dissolving 1.45 g of impure potassium hydroxide to make 250 cm^3 solution.
 - 1.3 Phenolphthalein indicator (P.O.P)

Procedures

- (i) Pipette 25 cm^3 of solution R into conical flask
- (ii) Add two drops of P.O.P indicator into conical flask
- (iii) Titrate solution O against solution R Obtain three more readings and record your results in a tabular form

Questions

- (a)
 - (i) The colour change was from ___ to ___
 - (ii) Calculate the average volume of acid used
 - (b) Find
 - (i) Number of moles of acetic acid used in the reaction
 - (ii) Number of moles of potassium hydroxide used in the reaction
 - (c) Calculate the molarity of acid (CH_3COOH)
 - (d) Calculate the percentage purity of potassium hydroxide
2. As a chemist working at Non Governmental laboratory department, you are required to identify cation and anion present in a salt sample W brought in your office. Carry out the following experiments and record your observation in a tabular form. Appearance, Action of heat on a sample, Action of dil HCl on a sample, Solubility, Action of NH_3 solution on a sample solution, Flame test and addition of MgSO_4 on a sample solution followed by warming the content.

CONCLUSION

- (a)
 - (i) Cation present is ___
Anion present is ___
 - (ii) Chemical formula of a sample is ___
Chemical name of a sample is ___
- (b) Explain the uses of salt sample W in daily life